Reading summaries

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Research summaries

Correlational data-driven belief models

During our meeting on December 17, Sara shared several papers related to data-driven models of internal belief systems (Brandt, 2022; Brandt et al., 2019; Lee et al., 2025; Powell et al., 2023; van Noord et al., 2025).

The studies describe attempts to infer belief systems from survey data, for the purposes of either proposing such a method or as a means to study properties of these belief systems. Table 1 outlines the key features of these studies. Most of the examined papers consider population-aggregate belief systems, and treat belief systems as correlational networks. The surveys cover a variety of belief system contexts, spanning general political attitudes and ideologies, climate change policies, as well as beliefs and intentions concerning childhood vaccination.

Only one of the studies, Powell et al. (2023), considers directional belief relations, in the form of a Bayesian Network. While the authors of this study employ structure learning to identify edge existence and weight, they restrict the search space by forbidding edges flowing from more specific to more general concepts[1]. All other studies model belief systems as either regularised partial correlation networks (Brandt et al., 2019; Lee et al., 2025) or a related correlational measure of belief association (Brandt, 2022; van Noord et al., 2025).

Most of the studies consider population-aggregated belief systems[2], for which edges are inferred from population-level associations observed in the survey data. These have limited interpretability at the individual level (Brandt & Morgan, 2022), e.g., for inferring structural variation across individuals' belief systems, or reasoning about interventions. The only examined study to explicitly consider individual belief systems is Brandt (2022), which investigates differences in belief system structure using individual conceptual similarity ratings[3]. In all other studies the relations between beliefs should be considered general trends or statistical associations.

None of the studies consider higher-order belief relations comprising three or more beliefs. This is a potential limiting factor on claims for model accuracy, in particular due to the multiple realisability of the survey data observations when higher-order interactions are permitted. That is to say that claims for the accuracy of the inferred models are subject to the assumption that all meaningful belief relations are pairwise.

Text	Model	Aggregation	Context	Research intent
(Brandt et al., 2019)	R-PCN	Population	General political	What *types* of beliefs are most central in belief networks: operational or symbolic?
(Brandt, 2022)	CSN	Individual	General political	Method to infer (undirected) individual belief systems.
(Powell et al., 2023)	BN	Population	Vaccination	Role of auxiliary hypotheses/beliefs in intervention outcome and effect on belief revision.
(van Noord et al., 2025)	R-PCN **(?)**	Cluster	General political	Method to identify and compare belief systems within/ between populations.
(Lee et al., 2025)	R-PCN	Country	Climate change	Analyse stability and inconsistency of climate belief systems across globe.

Table 1: Model, context, and intentions of (considered) papers on data-driven belief models. Models: **R**egularised **p**artial **c**orrelation **n**etwork (**R-PCN**), **C**onceptual **S**imilarity **N**etwork (**CSN**), **B**ayesian **N**etwork (**BN**). Belief systems aggregated at varying levels — non-individual aggregations to be thought of as general trends/statistical associations.

Brandt (2022) statistical control issues

We identified potential methodological issues in the choice of control variables for the statistical analysis in Brandt (2022). In this study the authors regress survey participants' conceptual association ratings for belief pairs on a number of individual difference factors (political identity, political engagement, political knowledge, liberal/conservative attitudes) to investigate their impact on belief system structure. While the authors do not

[1]Powell et al. remark that the relations should not *all* be interpreted causally, but rather as 'probabilistic dependencies or "influence" relations'. They evidence this point with respect to the inferred edge `Vaccine Danger` ⟶ `Vaccines Toxicity`; intuitively, people do not consider vaccines toxic *because* they consider them dangerous. Instead, the authors argue that this relation is more appropriately interpreted as set membership.

[2]These are also referred to as 'between-person' belief systems.

[3]The survey for this study asks participants to judge the conceptual association between pairs of beliefs or attitudes. For instance: 'Someone who believes in *X* is likely to also believe in *Y*.'

describie their assumed causal model for these analyses, at least one plausible model indicates problems in their choice of control variables, which could bias their results. We note that the authors appear to recognise the interdependencies of these variables, as they separate their statistical analysis into sets of control variables rather than performing a single linear regression (which would imply an independence assumption).

Let i refer to some survey participant, and p a pair of beliefs. The response variable is the individual's conceptual similarity rating for the belief pair ($R_{i,p}$). Individuals are characterised by:

I_i	**I**deological identification (liberal $\longleftrightarrow$ conservative)
X_i^I	**I**deological e**X**tremity
P_i	**P**arty identification (democrat $\longleftrightarrow$ republican)
X_i^P	**P**artisan e**X**tremity
E_i	Political **E**ngagement level
K_i	Political **K**nowledge

Ideological and partisan extremity factors are constructed from the corresponding identification factors by '"folding over" the measure' (Brandt, 2022, p. 838). Participants are assigned randomly to one of two experimental **C**onditions[4]: C_i. Pairs of beliefs are tagged with a **T**ype: T_p, according to whether the beliefs are ideologically consistent or inconsistent.

To construct a causal graph for conceptual similarity ratings from the described variables[5], we assume that:

1. Each of the stated factors could have a direct **causal effect** on an individual's conceptual rating for a given belief pair.
2. Ideological identification is a **causal factor** for party identification (e.g., people who view themselves as conservative are more likely to vote Republican on account of its general recognition as the conservative party).
3. Ideological identification is a **causal factor** for political engagement.
4. Political engagement is a **causal factor** for political knowledge.
5. Ideological extremity is a **causal factor** for partisan extremity (i.e., individuals who are firmly liberal/ conservative are likely to be also firmly partisan).

Furthermore, since the extremity factors are constructed, these are **causally dependent** on the corresponding identification factors by definition. The resulting causal DAG is shown in Figure 1.

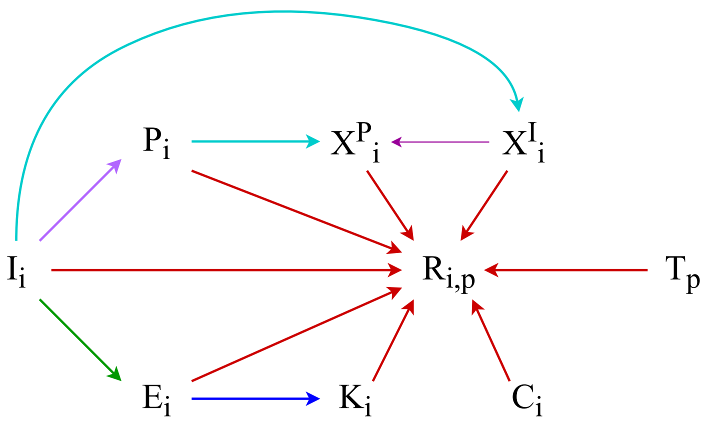

Figure 1: Causal DAG for conceptual similarity ratings ($R_{i,p}$) in Brandt (2022), where i represents a participant in the survey, and p is a belief pair. Edge colours correspond to causal assumptions outlined in main text. Factors are: ideological identification (I_i), partisan identification (P_i), ideological extremity (X_i^I), partisan extremity (X_i^P), political engagement (E_i), political knowledge K_i, pair type (T_p; ideologically consistent or inconsistent), and experimental condition (C_i; questions about *own* beliefs or *others'* beliefs).

[4]The experimental condition determined whether survey participants are asked to rate conceptual similarity of beliefs for themselves (e.g., "Imagine you hold belief *X*. How likely are you to believe *Y*?"), or for other people (e.g., "Suppose that someone holds belief *X*. How likely are they to believe *Y*?").

[5]We acknowledge that there may exist other factors which should be considered in this model, but do not consider these here since our intention is to illustrate a problem with the use of existing variables.

Brandt performs four statistical analyses on this set of variables. We will summarise the estimands, choice of controls, and correct controls for each one. In each case they separate the the analysis according to belief-pair ideological consistency (T_p) and experimental condition (C_i), controlling for both factors.

Importantly, we stress that this is one plausible DAG for which the discussed issues may be problematic. It is possible that the authors have different causal assumptions for which these controls are correct. Our critique is that unless these assumptions are made explicit, some results have ambiguous interpretation and significance.

1. $E_i \longrightarrow R_{i,p}$

The authors first investigate the effect of political engagement on rating. They control for E_i, but leave open a backdoor path through I_i (according to our assumed model in Figure 1). If political engagement varies with ideological identification then the measured effect will be biased. The correct control set to estimate the *total* effect is $\{E_i, I_i\}$. If the authors intend to measure the *direct* effect (i.e., excluding the portion of the effect due to influence of political engagement on political knowledge), they should also control for K_i.

2. $K_i \longrightarrow R_{i,p}$

They then estimate the effect of political *knowledge* on rating, controlling for K_i, leaving open a backdoor path through E_i. The correct control set is $\{K_i, E_i, I_i\}$. They observe a similar effect here as in the first study; however, since this includes the effect of political engagement this is not measuring the effect of interest.

3. $I_i \longrightarrow R_{i,p}$ **and** $X_i^I \longrightarrow R_{i,p}$

Thirdly, they investigate two estimands with the same model, i.e., the effects of ideological identification and partisan identification, controlling for I_i and X_i^I. These controls are correct for estimating the effect of X_i^I, but not for I_i.

It is unclear from the text whether the authors intend to measure the direct effect of ideological identification, or the total effect (i.e., including the intermediary causal paths through party identification, political engagement, etc.). If they intend to measure the *direct* effect, then the correct control set is $\{I_i, P_i, E_i, X_i^I\}$. If they intend to measure the *total* effect, then the correct control set $\{I_i\}$ excludes X_i^I.

4. $P_i \longrightarrow R_{i,p}$ **and** $X_i^P \longrightarrow R_{i,p}$

As above, they investigate two estimands with a single model, controlling for P_i and X_i^P. We treat these two estimands separately.

For partisan identification, the correct control set is $\{I_i, P_i\}$ for the *total* effect or $\{I_i, P_i, X_i^I, X_i^P\}$ for the *direct* effect. For partisan extremity the direct and total effects are equivalent, and are correctly estimated using the control set $\{P_i, X_i^I, X_i^P\}$.

Survey overlap

Topic	Studies (Brandt (2022))	Climate Attitudes waves
The Democratic party	1,2,3	1,2,3,4,5
The Republican party	1,2,3	1,2,3,4,5
Environmental protection	1,2	1,2,3,4,5
Gay rights	1,2	
Gun ownership	1,2	
Regulations on big business	1,2	1,2,3,4,5
Abortion rights	1	
Government aid for Black people	1	
Government run healthcare	1	5
Military spending	1	
Restrictions on immigration	1	1,2,3,4,5
Severe criminal sentences	1	4
Taxes on the rich	1	
Social security	2	?
Mandatory vaccines	2	5
Limits on medical malpractice suits	2	
Legalised marijuana	2	
Increased education funding	2	
Free trade with China	2	
Ban smoking in public places	2	
Requiring all students to receive the COVID-19 vaccination	3	
Increasing the number of Supreme Court justices	3	
Going to war with Iran	3	
Funding more research into renewable energy sources	3	2
Federal investments in infrastructure	3	1,2,3
Establishing an independent Palestinian state	3	
Deporting immigrants working in the United States illegally	3	
Banning police from using chokeholds	3	2,3,4,5 Similar (police brutality)
Background checks on all gun sales	3	
Allowing vote by mail in all elections	3	3
A wealth tax on wealth above $25 million dollars	3	

Table 2: Survey overlap between Brandt (2022) and Climate Attitudes dataset

Category	Belief topic	Further explanation	Climate Attitudes waves
Vaccines	Vaccination intentions	Intention to vaccinate children (or hypothetical children)	
	Vaccine danger		5
	Toxic additives in vaccines		
	Vaccine effectiveness	Effectiveness at *preventing* disease	5
Childhood diseases	Disease rarity		(Similar: threat to family) 1
	Disease severity		(Similar: threat to family) 1
Infant immune system (IIS)	IIS: weakness		
	IIS: limited capacity	Limited in capacity and easily overwhelmed	
	IIS: vaccines strain	Vaccines strain the infant immune system	
Parenting and medicine	Parental protectiveness		
	Parental expertise	i.e., that parents usually know more about their children's health than medical experts	
	Medical scepticism	Including concerns about pharmaceutical companies and corruption in the medical community	
Worldviews	Naturalism	Preference for natural over artificial things	
	Holistic balance	Related to attitudes toward alternative medicine	

Table 3: Psychometric scales used in a Study 1 of (Powell et al., 2023).

Belief topic	**Climate Attitudes waves**
Left-right identification	1,2,3,4,5
Gender inequality	
Anti-LGBT	
Euroscepticism	
Anti-immigration	2,3,4,5
Anti-egalitarianism	2,3,4,5
Benefits harm economy	
Benefits harm society	1,2,3,4,5
Welfare chauvinism	1,2,3,4,5 (indirect)
Anti-economic interventionism	5
Anti-social benefits (low income)	coincides with “Benefits harm society”
Anti-social benefits (parents)	
Educational spending	
Anti-basic income	3
Anti-climate change taxes	1,2,3
Anti-climate change renewables	2
Anti-climate ban of appliances	
Climate scepticism	1,2,3,4,5
Authoritarianism	1,2,3,4,5
Anti-libertarianism	(Unclear distinction with authoritarianism)

For Lee et al. (2025), I believe all variables are represented in the climate attitudes survey.

Also consider the nature of the questions asked by each study. What would these look like in a climate change context? What variables do we have that fit the description?

Partial correlation

Kyuri recommends starting with (Epskamp et al., 2018b) and possibly (Epskamp et al., 2018a).

Papers which use partial correlation in belief/internal cognitive state context:
- (Brandt et al., 2019)
- (Fishman & Davis, 2022)
- (Lee et al., 2025)
- (Brandt & Sleegers, 2021)
- (Dalege et al., 2025)
- (Dalege & Van Der Does, 2022)

Other papers which discuss partial correlations:
- (Park et al., 2024)
- (Epskamp et al., 2018b)
- (Epskamp et al., 2018a)
- (Borsboom et al., 2021)
- (Yin & Li, 2011)

Criticism:
- (Brandt & Morgan, 2022)

The Gaussian Graphical Model in Cross-Sectional and Time-Series Data (Epskamp et al., 2018a)

Let Σ be a variance-covariance matrix with rank n, then the precision matrix is defined as the inverse, $K = \Sigma^{-1}$, if it exists. Standardising K gives the partial correlation matrix:

$$R = \mathrm{Cor}\left(Y_i, Y_j \mid \boldsymbol{y}_{-(i,j)}\right) := -\frac{\kappa_{ij}}{\sqrt{\kappa_{ii}.\kappa_{jj}}} \tag{1}$$

Note that the diagonal elements equal 1 by definition, rather than by this formula, and should not be negated.

Causal interpretation of edges

An edge $A - B$ appears in a GGM only if there is a causal link $A \to B$ or $B \to A$, or a collider $A \to C \leftarrow B$ which we condition on. The authors do not consider the possibility of an unobserved fork, likely because they assume that "a causal model between *observed* variables generated the data".

So non-zero partial correlations/edges in a GGM can be considered indicators of potential causal relations involving the variables of interest. However, causal directionality, and the presence of forks or colliders is underdetermined. In the case of forks, unobserved common causes are indistinguishable from direct relations ($A \leftrightarrow B$), while observed forks are conditioned on and thus reduce the observed relation between A and B. With regards to colliders, the implications are reversed. Unobserved colliders do not affect the partial correlation matrix since they are not conditioned on, and thus may not be detected. Observed colliders bias the relation between the causes, and thus feature as a modified direct relation.

Question: Why do we observe a non-zero partial correlation between A and C in the fork $A \leftarrow C \to B$? To calculate this, we control for B. However, since A and B are only caused by C, I would expect the partial correlation then to be zero, since we remove the comparable relation between C and B.

Estimating Psychological Networks and their Accuracy: A Tutorial Paper (Epskamp et al., 2018b)

Problem: Use of psychological networks linking psychological behaviour to underlying psychological factors is popular, but evaluation measures have not been well-explored.

Contributions:
- Introduction to SOTA psychological network estimation
- Discussion on importance of investigating their accuracy

- Provide a methodology for evaluating accuracy, stability (of centrality indices), and comparing variables in psychological networks.
- Introduce novel statistical methods:
  - Correlation stability coefficient
  - Bootstrapped difference test
- Produce R package (`bootnet`) for estimating and evaluating psychological networks using proposed measures.
- Case study using `bootnet`.

The supplementary material includes description of how to interpret psychological networks, and overview of network measures.

(Van Borkulo et al., 2014)

Critical reflection

What is Central to Political Belief System Networks (Brandt et al., 2019)

Audience: Quantitative political science and sociology

Problem: Prior attempts to determine central components of political belief systems do not typically consider the global network structure of belief systems, so may falsely identify peripheral relationships between beliefs/ attitudes as central.

Research questions:
- Are operational components of belief systems more 'central' in those systems than symbolic components, or vice versa?
- To what degree are operational and symbolic components associated with political (voting) behaviours? (Based on the hypothesis that more central ⟶ stronger association.)

Claims/contributions:
- (Contribution 1) Partial correlation network constructed for operational and symbolic components of political belief systems, based on longitudinal survey data from New Zealand.
- (Contribution 2) A more continuous method than is typical for assessing centrality of belief system components. Authors assert that typical approaches dichotomise components into 'central' and 'peripheral' categories.
- (Claim 1) Symbolic components are more 'central' than operational components, on all considered measures of centrality (strength, closeness, betweenness).
- (Claim 2) If symbolic or operational components tend to be more 'central' than the other, then the more central component should be closer in the network to voting behaviour.
  - *This claim should be examined more closely. While they cite other work as supporting it, the claim is not well-reasoned in the paper, especially given the loose treatment of 'centrality' interpretation in partial correlation networks.*
- (Claim 3) In political belief systems, symbolic attachments to parties and labels are more important than actual policy positions.
  1. *Does the strong importance claim follow from the reported centrality results, i.e., is their measure of centrality sufficient to encompass the typical dimensions we may judge the importance of a belief system component?*
  2. *How sensitive are the centrality results (and hence importance claim) to the authors' use of a partial correlation network?*

Other notes:
- The study uses longitudinal data primarily to check consistency of results over time. i.e., they construct a partial correlation network for each wave of survey data.
- The authors acknowledge the limited interpretability of centrality in absence of causation (their edges are directed partial correlations), and that causal structures could theoretically change the support for their claims. In particular they refer to the possibility of a sink-like structure where some belief is only a consequence, and never a cause; however, *they consider this structure unlikely to occur.*

Operational and symbolic components: Operational components refer to particular positions on issues, while symbolic components refer to 'affective attachments to political groups and labels'. The authors identify survey questions deemed as assessing each of the categories. A question is considered **symbolic** if it asks about support (identification) for (with) political groups or labels. A question is considered **operational** it is concerns support for a particular policy. Overall there were

Possible limitations:

- The term 'centrality' can be used generally in multiple different senses, but has specific technical meanings in network science. The authors have provided some justification for their centrality measures with respect to dynamics on the partial correlation network, though I find this falls short in two ways. Firstly, they do not justify that these measures capture the typical (everyday) uses of the term 'centrality' in this context. Secondly, they provide limited discussion on interpreting partial correlation networks as belief systems. Thus the use and interpretation of these measures have not been well-justified.

- The authors consider only binary relations between components.
- The belief system components are not beliefs per-se, but rather positions or attitudes implied by underlying beliefs. For instance, consider the survey question regarding '*Reserving places for Māori students to study medicine*'. Participants may not have previously formed any particular belief regarding this issue, yet their response may depend on their established beliefs concerning equitable societies and the importance of medicine as a field of study. Furthermore, it is conceivable that this causation flows in one direction, i.e., that an individual's position on this issue cannot measurably influence their underlying beliefs. If this is true, and measurements have unmeasured common causes, then correlation between responses may not imply any direct relationship between these components.
- Some of the centrality measures require a different interpretation in directed networks. For instance, strength and closeness may vary when considering incoming versus outgoing edges.
- The measurement interval (2009-2016) does not include a change in government.
- The networks do not attempt to describe belief systems of individuals, but rather variation across the population.

Survey data:
- Longitudinal survey (NZAVS) with seven waves; representative sample of New Zealand.
- Items are assessed as either symbolic or operational.
- Symbolic items were: "support for party X", ideological identification, and right-wing/left-wing identification.
- Operational components all query participants support for a particular existing or potential policy, with responses coded such that higher scores represent more conservative positions in NZ context.

Measuring the belief system of a person (Brandt, 2022)

Audience: Political belief systems/sociology

Problem: Typical empirical studies on belief systems consider data aggregated at the population level, while theories of belief systems assume these to be individual-level constructs. In other words, such empirical studies violate a core assumption of their underlying theory.

Research questions: To what degree are conceptual similarity reports a reliable and valid approach to estimating individual belief system structure in a political belief context?

Claims and contributions:
- (Contribution) Conceptual similarity as a method for estimating individual's belief system structures.
- (Claim) Conceptual similarity ratings accurately track true similarity between political attitudes
- (Claim) Conceptual similarity observations are consistent with behaviour predicted by theoretical computational models (Brandt-Sleegers, Goldberg-Stein)
  - **Note:** This is subject to the statistical controls limitation discussed below.

Limitations
- Possible issues in statistical controls may invalidate some results from studies 1—3
- Multiple realisability may compromise ability to draw structural conclusions from pairwise belief survey questions (cf. Guest)
- Conceptual similarity does not capture causal directionality
- They classify attitude pairs as ideologically consistent or inconsistent. However, this (i) assumes a single dimension for ideological, and (ii) dichotomises each pair, when we could reasonable speak of degrees, and expect differing degrees to affect dynamics.

Connections to other research
- Reflecting on two theoretical/computational models of belief systems (Brandt-Sleegers, Goldberg-Stein).

Possible error in choice of Controls (Notes, causal DAG on iPad)

Survey data:
- Five studies

- Measures of participants' attitudes: support/opposition for 12 salient political positions, coded such that higher scores reflect more conservative attitudes
- Political engagement: "How interested in politics are you?", "How important are politics to you?"
- Political knowledge: 10 multiple choice factual questions about current events, political power, and government procedures.
- Conceptual similarity (following methods from Stolier et al. (2020): either asked as "own-task" or "other-task" condition. Participants either rated both orders of an attitude pair (study 1), or a random selection (studies 2 and 3).
  - Own-task: "Imagine that you support one attitude, how likely is it that you will support the other?"
  - Other-task: "Given that a person supports one attitude, how likely is it that they support the other?"
- Demographic information: age, income, education, ethicity/race, English language, state, religious affiliation, gender.
- Ideological identification: "When it comes to politics, do you think of yourself as a liberal, conservative, moderate, or haven't you thought much about this?"
- Partisan identification: "Do you think of yourself as a Republican, a Democrat, an Independent, or haven't you thought much about this?"

Modelling and leveraging intuitive theories to improve vaccine attitudes (Powell et al., 2023)

Overview: Per the Duhem-Quine underdetermination thesis, when an individual's beliefs on a particular topic are put to question (e.g., after observing relevant evidence, or following a targeted intervention), the degree and direction in which belief revision happens can be sensitive to auxiliary hypotheses or beliefs, which conflict or reinforce the target belief.

The authors propose a method using Bayesian networks to model intuitive theories. They select 14 beliefs judged as related to a selected target belief (vaccination intention), and undertook a survey to determine individual positions on each of these beliefs. Each belief was assessed using several survey questions, with the final value taken as the normalised average score. The relations between beliefs were inferred using a Beta regression on the weights between beliefs (as conditional probabilities), with the added prior restriction that directionality should flow from more general to more specific beliefs. This model captures conditional probabilistic relations between beliefs, at a *population-aggregated* scale.

Using the fit model, the authors perform several experiments testing its adequacy at explaining the effects of belief intervention, and observation of evidence, on both the target belief and auxiliary beliefs. This includes replicating an existing study shown to be effective at shifting vaccination intentions, and a novel study for the same purpose, where the authors instead intervened on a belief predicted by the model to be promising for intervention.

Audience: Cognitive modelling, applied and theoretical psychology

Problem: Intuitive theories, which support much of human thought, can also reinforce misconceptions which constitute a major health risk in the context of vaccination. To address these misconceptions, it is necessary to understand how they are produced by underlying conceptual/belief systems. Yet our understanding here is limited.

Research questions:
- To what degree is belief-revision (from intervention; from observation of evidence) sensitive to auxiliary beliefs? (cf. Duhem, Quine)
- Can inferred population-aggregate-level Bayesian networks representing intuitive theories be used to non-trivially (i) aid in explaining the success or failure of belief revision outcomes, (ii) reason about the effects of intervention on distant, or indirectly-related beliefs, and (iii) design novel effective interventions for vaccination intention?

Claims and contributions:
- (Contribution) A method for generating directional cognitive models of belief (of a population) for a particular domain.
- (Contribution) A plausible population-level cognitive model of beliefs relating to vaccine intention.

- (Claim) Domain-specific Bayesian network models of population-aggregated beliefs can help explain the success or failure of interventions, with respect to auxiliary hypotheses or beliefs.

Limitations:
- Model is population-level, so aggregates over individuals rather than representing individual differences.
- "Difference in difference" analyses are difficult to interpret, since they do not incorporate distributional skew.

Connections to other research:
- Authors note limitations of structure learning algorithms using correlations to identify causal directionality. They suggest future work could ask participants about their perceived causal directionality. This is similar to (Brandt, 2022).

Study intentions:
1. Develop a cognitive model of 14 beliefs related to vaccine hesitancy in U.S.
2. Replicate success of an existing intervention, and use model from (1) to explain *how* and *why* the intervention works, as well as its effects on related beliefs.
3. Create and validate the success of a novel intervention to address concerns about toxic additives in vaccines.
4. Test whether peoples' beliefs changed coherently following relevant real-world events, examining beliefs before and after a serious measles outbreak.

Other notes:
- They show that intervening on one part of the belief network can induce changes in other parts. Perhaps this could also inform belief system inference. If we intervene on a particular belief and another belief changes, then they are likely closely related in the network. If some other belief shows no change, then the two are likely not related, related only weakly, or only via a higher-order interaction, e.g., with a third belief.
- They note that auxiliary hypotheses and beliefs can affect belief revision outcomes (per Duhem-Quine thesis).

Survey data:
- Assesses 14 beliefs/attitudes/intentions (Table 5).
- Each belief assessed using 4-6 statements, including at least one reverse-coded item. Scale items listed in supplementary materials.

Category	Belief topic	Further explanation
Vaccines	Vaccination intentions	Intention to vaccinate children (or hypothetical children)
	Vaccine danger	
	Toxic additives in vaccines	
	Vaccine effectiveness	Effectiveness at *preventing* disease
Childhood diseases	Disease rarity	
	Disease severity	
Infant immune system (IIS)	IIS: weakness	
	IIS: limited capacity	Limited in capacity and easily overwhelmed
	IIS: vaccines strain	Vaccines strain the infant immune system
Parenting and medicine	Parental protectiveness	
	Parental expertise	i.e., that parents usually know more about their children's health than medical experts
	Medical scepticism	Including concerns about pharmaceutical companies and corruption in the medical community
Worldviews	Naturalism	Preference for natural over artificial things
	Holistic balance	Related to attitudes toward alternative medicine

Table 5: Psychometric scales used in a Study 1 of (Powell et al., 2023).

The nature and structure of European belief systems: exploring the varieties of belief systems across 23 European countries (van Noord et al., 2025)

Overview: The authors present a 'bottom-up' method for identifying (population-aggregate) belief systems within and between populations, based on shared correlational structures rather than particular belief states. They use this method to analyse existence and similarity of belief systems across Europe.

Audience: Social psychology, cognitive modelling

Problem: Belief system research has predominantly focused on single countries, and often the US. We have limited insight into whether and how belief systems vary between countries.

Intent:
- What do belief systems in European societies have in common?
- What do the people who hold different belief systems in European societies have in common?

Research questions:
- Are political belief system structures similar *across* Europe?
- Which demographic groups are likely to have similar belief systems *within* countries?
- How are belief systems related to voting behaviour?

Methods:
- *Identifying belief systems:* Correlational class analysis (CCA) per country to identify **clusters of individuals** based on how their beliefs relate to one another (i.e., they may not hold the same beliefs, but their beliefs are correlated in the same ways). The clusters represent belief systems.
- *Structure of a belief system:* For each correlational class (cluster of individuals), authors produce a correlation matrix relating beliefs to beliefs. This can be viewed as a network, where strong positive (negative) edges reflect strong positive (negative) correlations between beliefs.
- *Identifying similar belief systems:* Authors calculate Pearson correlation between each pair of belief systems, across all countries. The resulting correlation matrix reflects the similarity between each pair of belief systems. Authors cluster the corresponding network to identify groups of similar belief systems.
- *How many dimensions per belief system:* PCA on each belief system separately. Then factor analysis to determine possible number of "meta-dimensions" that organise the identified principal components.
- *What are the underlying dimensions:* Finding little support for a 1D belief-system, authors consider the two-factor solution to factor analysis (above), counting how often two beliefs have > 0.3 loading on the same dimension (factor).
- *How are the dimensions related to one another:* **Didn't quite understand, as it relied on understanding the prior analysis classifying the underlying dimensions, and I did not quite get this.**

Findings:
- *Number of belief systems:* 70 across 23 countries, with 2—5 per country.
- *Groups of systems:* Two clusters of belief systems.
- *Similarity between belief systems:* Across all countries, between-belief system correlation is in $[-0.053, 0.792]$ with a mean of 0.326. So some similar belief systems, but lots of variation.
- *Within-cluster similarity:* Group 1 more similar than Group 2.
- *Constraint (structuredness):* Similar mean density between the groups. Within each group there exists more variation in density. Key finding: main difference between groups is not their structuredness.
- *How many dimensions per belief system:*
  - PCA: Between 6 and 9 per belief system with eigenvalues > 1.
  - Factor analysis: Almost all belief systems in Europe multidimensional.
- *What are the underlying dimensions:* Belief-pairs with high loading (> 0.3) on a given factor typically both economic or both cultural. Suggests factors describe economic or cultural dimensions rather than a mix.
  - **Struggled to understand this section — to review**
- *How do dimensions relate to each other:* In Group 1 right-wing cultural beliefs tend to go with right-wing economic beliefs, while in Group 2 right-wing cultural beliefs tend to go with left-wing economic beliefs. However, the

correlations are small. Likely due to many factors not classified as cultural or economic — if these follow a different logic then their correlations are unlikely to be in-line with cultural or economic factors.

Claims and contributions:
- (Contribution) A 'bottom-up' method for analysing the belief systems held *within* a population and *between* populations.
- (Contribution) Qualitative analysis of the variation in, and types of belief systems present across Europe.
- (Claim) Belief systems in Europe can be broadly categorised into two groups, one of which exhibits positive correlation between cultural and economic dimensions (i.e., right-wing goes with right-wing), and one of which exhibits negative correlation. These types were also associated with various demographic features (in particular **education**), and geographic location in Europe.

Limitations:
- Only considers bidirectional correlations
- Some potential statistical methodological issues in analysis (treating demographic variables as independent; excluding data with 'zeroes' rather than modelling it)

Connections to other research:
- Mark Brandt is a co-author, and authored several other papers we consider here (Brandt, 2022; Brandt et al., 2019).
- Cites Converse, like many of the other papers we have considered.

Other notes:
- Survey questions selected from the 2016 European Social Survey, comprising question to test both operational and symbolic components of belief systems.

Survey data:
- 20 political beliefs.
- Mostly single-items, some scales (averaged across items).
- Both operational and symbolic components.
- Broad set of beliefs, not all easily categorised as 'cultural' or 'economic'.
- Coded such that higher scores indicate more right-wing responses.
- Topics span: ideological identification, gender + LGBT, EU, immigration, role of government, social benefits, education spending, climate change (taxes, subsidising renewables, banning low-efficiency appliances, scepticism), authoritarianism, libertarianism.

Variations in climate change belief systems across 110 geographic areas (Lee et al., 2025)

Overview: The authors infer population-aggregate climate change belief systems for countries around the globe, modelled as regularised partial correlation networks. They analyse differences in belief system stability (density) and inconsistency, to illuminate possible areas for belief intervention. They find geographic variation in density (higher in the global north), and inconsistencies particularly relating to fossil fuel usage. Density is positively correlated with education and GDP, while inconsistency correlates positively with carbon resource rent and negatively with education.

Audience: Cognitive modelling, social psychology

Problem: Population-level climate action requires individual belief systems to be aligned with this. Conflicting beliefs or doubt about climate change can hinder this. Furthermore, the structure and resilience of belief systems can affect the effectiveness and outcome of interventions. However, there is limited research on these properties, and even less considering possible geographical variation. This limits our ability to understand the effects of intervention, and how these may vary in different contexts or locations.

Intent: Identify cross-societal differences in the stability and coherence of climate change belief systems around the globe.

Research questions:
- How does density (reflecting stability) of belief systems vary by country, global north/south divide, and other societal factors (below)

- How does inconsistency (reflecting belief-pair conflicts) of belief systems vary by country, global north/south divide, and other societal factors (below)

Additional factors: education level, exposure to climate change information, GDP, carbon resource rent.

Methods:
- A belief system for each area was estimated using eight key climate-related beliefs and attitudes commonly assessed in public surveys.
- Network edges were determined through partial correlations among these elements.
- Uses proportional measures of belief system density and inconsistency.
- Data from survey of Facebook users (n=99,074)

Claims and contributions:
- Countries in the global north exhibit more interconnected belief systems (more large edges) than those in other regions.
- Density varies significantly by region; typically higher in global north, reflecting more stable belief systems.
- Level of interconnection correlates negatively with pro-climate beliefs and attitudes. i.e.,
  - More stable (population-level) belief systems are typically more anti-climate, typically in the global north.
  - Low density, pro-climate is more typical in the global south, reflecting supportive but less stable systems.
- Inconsistencies center around fossil fuel use, and:
  - Renewable energy support (Nigeria, Zambia, the Philippines, and Jordan), suggesting support for renewable energy use, but not for reducing fossil fuel usage.
  - Support for governmental climate priority (Laos, Kuwait, Vietnam, Jordan), suggesting support for the government making climate change a priority, but not for reducing fossil fuel usage.
- Belief system stability (density) is positively correlated with information exposure relating to climate change, and GDP. It is not correlated with education or carbon resource rent.
- Belief system inconsistency is correlated *negatively* with information exposure relating to climate change, and *positively* with carbon resource rent. Suggests higher economic dependence on carbon resources begets more inconsistency.

Limitations:
- Population-level aggregate belief-systems.
- No causal model in analysis of factors, though some could reasonably be hypothesised to be causally related (e.g., education and information exposure).
- Their 'inconsistency' measure is taken as the absolute proportion of negative correlations. This assumes all beliefs are coded such that high or low values all correspond to the same situation (pro- or anti-climate). Hence this measure only measures inconsistency with respect to the variable of interest, and is not applicable when beliefs may conflict on other dimensions (e.g., cost, social impact, ideology).

Connections to other research:
- Cross-regional variation in belief systems also considered for European nations in (van Noord et al., 2025), though they were more concerned with meta-level similarity and relationships between belief systems, and qualitative 'dimensions', than structural properties.

Survey data:
- International study of Facebook users from 110 geographical areas ($n = 99,074$).
- Eight climate-related beliefs commonly assessed in public surveys, characterised as beliefs, risk perceptions, or policy support.
- Beliefs: climate change is happening; assuming CC happening, it is caused by human activities.
- Risk perception: concern about CC; personal harm expected from CC; future generation harm from CC.
- Policy support (for country where you live): in future should use more/less/same amt fossil fuel; in future should use more/less/same amt renewable energy sources; CC should be very high/high/medium/low priority for government.

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